

PROJECT DOCUMENTATION

US E-Commerce Analytics

From raw CSV extracts to a verified Power BI semantic model, a five-page dashboard, and the business insights one million e-commerce orders support in a three-layer SQL Server warehouse, documented end to end.

End-to-End Data Warehouse & BI
SQL Server 2025 · T-SQL · Power BI
Dataset 2019–2025 · 1,000,000 orders
Verified 2026-09-26

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1. Executive Summary

Item	Detail
Project name	US E-Commerce Analytics — End-to-End Data Warehouse & BI
Objective	Build a production-grade analytical warehouse over 1M e-commerce orders and deliver a governed Power BI dashboard answering revenue, profitability, customer, operations, payments, and experience questions
Business problem	Raw order exports cannot answer business questions directly: revenue, payments, reviews, and delivery live at different grains; customer identity exists at two levels (account vs person); canceled orders carry structural NULLs; one reference table required deduplication
Dataset	Synthetic U.S. e-commerce dataset — 1,000,000 orders (per project brief); author/source/license: Not Available / Not Verified
Time period	2019-01-01 → 2025-12-31 (purchase timestamps; delivery/review timestamps spill into early 2026)
Records	8 source files, ~857 MB, 6,296,938 raw rows across 8 entities
Technology stack	SQL Server 2025 Developer Edition, T-SQL, SSMS, sqlcmd, Power BI Desktop (Import mode), draw.io, Flask + Chart.js (web companion)
Architecture	CSV → Bronze (raw landing) → Silver (cleansed/conformed) → Gold (Kimball fact constellation: 4 facts + 6 dimensions) → validation suite (80 checks) → Power BI semantic model → 5-page dashboard
Analytical objectives	Revenue/profit drivers, category & product performance, discount effectiveness, customer loyalty (delivered-based repeat analysis), geographic concentration, delivery performance and its relationship with satisfaction, payment/financing behavior, seasonality
Final deliverables	Re-runnable SQL pipeline (bronze/silver/gold), 80-check validation suite, verified Power BI semantic model (24 measures, 18 relationships), 5-page dashboard, this documentation, LinkedIn overview deck, web companion dashboard

The pipeline in one line: raw CSVs are landed unchanged in Bronze, cleansed and conformed in Silver, reshaped into a Kimball fact constellation in Gold, proven by an 80-check validation suite, and consumed by a Power BI semantic model whose 24 measures reconcile to the SQL layer to the cent.

Business value: the warehouse resolves grain conflicts, identity ambiguity, and data-quality issues *once, in the right layer*, so every KPI on the dashboard is computed from one trusted, tested model — and every number on every page can be traced back to the source files.

2. Business Problem & Objectives

Why the project was created

Raw order exports cannot answer business questions reliably:

- Revenue, payments, reviews, and delivery data live at **different grains** (line item, order, payment transaction, review) — naive joins fan out and corrupt sums.
- Customer identity exists at **two levels**: `customer_id` (one account per order placement) and `customer_unique_id` (the natural person). Conflating them inflates customer counts ~3.6×
- Canceled orders (66,252) carry **structural NULLs** in delivery columns — naive averages and joins silently distort results.

- One reference table (geolocation) held ~12.8 duplicate coordinate rows per ZIP — naive joins fan out revenue ~12.8×
- No governed layer existed between the CSVs and any reporting surface.

Business decisions the analysis supports

- Where to focus pricing and procurement (category margin analysis)
- Whether discounting drives volume (discount-band analysis)
- Which customer segments and age groups to invest in
- Peak-season capacity planning (November/December)
- Delivery SLA management (the 15-day satisfaction cliff)
- Payment-rail and financing strategy
- Cancellation monitoring (rate and value)

Requirements hierarchy

1. **Business requirements:** a trustworthy, refreshable analytics product covering sales, profitability, customers, geography, fulfillment, payments, and customer experience for 2019–2025.
2. **Analytical questions:** ten core questions (revenue trend, category revenue/profit, top products, AOV, discount effectiveness, repeat purchasing, segments, seller performance, delivery-vs-satisfaction, review trends) — see §12.
3. **Metrics/KPIs:** 24 governed DAX measures with locked definitions (§11).
4. **Insights:** quantified, evidence-linked findings (§16–§17), separated from observations and interpretations.

3. Dataset Documentation

- **Dataset name:** Synthetic U.S. E-Commerce Dataset — 1M Orders
- **Source / author / license:** Not Available / Not Verified (synthetic dataset provided for the project; no external documentation file was supplied)
- **Synthetic:** Yes. Patterns are internally consistent (financial identities reconcile exactly) but distributions are smoother than real-world data.
- **Time period:** purchases 2019-01-01 01:21:03 → 2025-12-31 23:59:31; delivery and review timestamps extend into early 2026.

Source File	Grain	Rows	Key	Purpose
customers.csv	one customer account per order placement	1,000,000	customer_id	Two-level identity: customer_unique_id = natural person (279,199 persons)
geolocation.csv	one coordinate point per ZIP entry (no natural key)	11,500	none	ZIP → city/state/lat/lng reference
orders.csv	one order header	1,000,000	order_id	Status + purchase/approval/carrier/delivery/estimated timestamps

Source File	Grain	Rows	Key	Purpose
order_items.csv	one line item within an order	2,199,819	order_id + order_item_id	Transaction price (post-discount), freight, discount rate, product, seller
order_payments.csv	one payment transaction	1,149,371	order_id + payment_sequential	Payment type, installments, value
order_reviews.csv	one review	933,748	review_id	Score (1–5), comment title/message, timestamps
products.csv	one product	2,000	product_id	Category, brand, dimensions, unit cost, list price
sellers.csv	one seller	500	seller_id	Seller company and contact attributes

Important structural characteristics (verified):

- `customer_id` is unique per order placement (1:1 with orders); `customer_unique_id` is the person (279,199 persons $\approx$ 3.6 accounts/person).
- Order items carry the **post-discount transaction price**; `products.price` is the **list price** (verified: zero-discount lines match list price exactly; discounted lines match $\text{list} \times (1 - \text{rate})$ rounded to cents).
- An order can span multiple sellers (seller lives on the line, not the order).
- An order can have multiple payment transactions (`payment_sequential` 1–2).
- Reviews exist only for delivered orders (933,748 reviews = 933,748 delivered orders).
- Canceled orders (66,252) have NULL delivery timestamps — structural, not a quality defect.
- Payments reconcile with order invoices: per-order $\text{SUM}(\text{payment_value}) = \text{SUM}(\text{price} + \text{freight_value})$ with **zero mismatches on all 1,000,000 orders**.

4. Data Architecture

```

8 CSV files (~857 MB)
  | BULK INSERT (TRUNCATE + full reload, typed columns, no constraints)
  ▼
BRONZE – raw landing layer: what the source sent, nothing else
  | INSERT...SELECT: TRIM, standardization, type tightening, dedupe, PKs, CHECKs
  ▼
SILVER – cleansed + conformed: one trusted row per source entity
  | INSERT...SELECT: surrogate keys, grain separation, business rules
  ▼
GOLD – Kimball fact constellation: 4 facts + 6 dimensions (the analytical contract)
  | Import mode (Power BI) + live SQL (web companion)
  ▼
SEMANTIC MODEL – 24 measures, 18 relationships, all single-direction
  ▼
DASHBOARD – 5 pages, question-titled visuals
  ▼
BUSINESS INSIGHTS – this documentation, §16–§17

```

Why each layer exists:

- **Bronze** is the recovery point: if any transformation is ever wrong, everything is re-derivable without re-touching the CSVs.

- **Silver** solves data quality and identity *once*: trimming, standardization, constraint enforcement, and the one intentional grain change (geolocation deduplication).
- **Gold** solves *analytics*: grains are re-declared as facts, entities become dimensions, surrogate keys replace business keys, and business rules (delivery bands, on-time flags) are computed once.
- **Power BI** consumes only Gold — never Bronze or Silver — so every dashboard number has one definition and one source.

5. Bronze Layer

Loading method: TRUNCATE TABLE + BULK INSERT per table — full reload, re-runnable by design.

Working ingestion configuration (verified on SQL Server 2025):

```
BULK INSERT bronze.customers
FROM '\data\customers.csv'
WITH (
    FIRSTROW = 2,          -- skip header
    FORMAT = 'CSV',
    FIELDQUOTE = '"',
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '0x0a', -- source files use Unix LF endings
    TABLOCK
);
```

Ingestion decisions:

- **Typed columns** (NVARCHAR/INT/DECIMAL(10,2)/FLOAT/DATETIME) rather than all-text: the source is a known, clean extract; typing makes bad data fail loudly at the door.
- **No constraints, no indexes** — correct for a raw landing layer.
- **No metadata columns** (source_file, load_timestamp, batch_id) — a deliberate decision: with a single static source and full reloads they carry no operational value. Documented here rather than silently omitted.
- **ROWTERMINATOR discovery:** '\n' fails on this SQL Server build; the hex literal '0x0a' is required because the CSVs use Unix LF endings.
- **Files are UTF-8, LF-terminated, no embedded quotes** — verified before load.

Bronze tables:

Table	Grain	Rows	Columns (types)	Source file
bronze.customers	customer account	1,000,000	customer_id, customer_unique_id, customer_name (NVARCHAR(200)); gender (NVARCHAR(200)); age (INT); zip, city, state, segment (NVARCHAR(200))	customers.csv
bronze.geolocation	coordinate point	11,500	zip_code_prefix (NVARCHAR(10)), lat/lng (FLOAT), city, state (NVARCHAR(20))	geolocation.csv
bronze.orders	order	1,000,000	order_id, customer_id (NVARCHAR(100)); status (NVARCHAR(50)); 5 timestamps (DATETIME)	orders.csv

Table	Grain	Rows	Columns (types)	Source file
bronze.order_items	order line	2,199,819	order_id, order_item_id, product_id, seller_id (NVARCHAR(100)); shipping_limit_date (DATETIME); price, freight_value, discount_rate (DECIMAL(10,2))	order_items.csv
bronze.order_payments	payment transaction	1,149,371	order_id (NVARCHAR(100)); payment_sequential (INT); payment_type (NVARCHAR(50)); installments (INT); payment_value (DECIMAL(10,2))	order_payments.csv
bronze.order_reviews	review	933,748	review_id, order_id (NVARCHAR(100)); review_score (INT); comment title (NVARCHAR(50)), message (NVARCHAR(200)); 2 timestamps (DATETIME)	order_reviews.csv
bronze.products	product	2,000	product_id, category, name, brand (NVARCHAR(100)); weight/length/height/width (INT); cost, price (DECIMAL(10,2))	products.csv
bronze.sellers	seller	500	seller_id, company, contact name (NVARCHAR(100)); contact gender (NVARCHAR(20)); contact age (INT); zip (NVARCHAR(10)); city, state (NVARCHAR(50))	sellers.csv

Known data-quality observations at Bronze (profiled, not fixed here — Silver's job): NULL delivery timestamps on canceled orders (structural); gender coded M/F; geolocation duplicated per ZIP; discount_rate concentrated on ~8.6% of lines.

6. Silver Layer

Pattern: TRUNCATE + INSERT...SELECT, one batch per table, dwh_create_date DATETIME2 DEFAULT GETDATE() audit column added to every table. Fail-loud NULL policy: key/measure columns are NOT NULL and no COALESCE defaults exist — bad source rows abort the load instead of entering the warehouse.

Transformations actually implemented:

Transformation	BEFORE → AFTER	Where
Text trimming	customer_name with padding → trimmed value	all text columns, all tables
Gender standardization	M/F → Male/Female, anything else → 'n/a'	customers, sellers
ZIP type alignment	geolocation zip INT → NVARCHAR(10)	geolocation
Geolocation deduplication	11,500 coordinate rows → 900 rows, one per ZIP prefix (AVG(lat), AVG(lng) centroid, MAX(TRIM(city)), MAX(TRIM(state)))	geolocation — the only grain change in Silver
Constraint enforcement	8 primary keys (2 composite), 14 CHECK constraints (price > 0, freight ≥ 0, discount 0–1, product dimensions > 0, ages 18–100, 4 order date-sequence checks NULL-tolerant for canceled orders)	all tables
Structural NULL preservation	canceled orders keep NULL delivery timestamps (66,252 rows)	orders

Transformation	BEFORE → AFTER	Where
Audit column	dwh_create_date DEFAULT GETDATE()	all tables

Row-count reconciliation (Bronze → Silver): identical on all tables except geolocation (11,500 → 900 — the documented deduplication).

Why the geolocation deduplication lives in Silver: the analytical model needs exactly one row per ZIP; the raw detail stays in Bronze so the transformation remains reversible and auditable.

7. Data Quality & Validation

The validation suite is `sql/validation/validate_gold.sql` — read-only, re-runnable, **80 checks in 8 categories**, each returning PASS/FAIL against hardcoded verified baselines. It is the gate before every Power BI refresh. **Current state: 80 PASS / 0 FAIL.**

Category	Checks	What it proves
Layer row counts vs verified baselines	26	bronze = silver = gold row-for-row (except the documented geolocation dedupe)
Gold grain (rows = distinct keys)	10	every fact/dimension PK is actually unique — catches duplicates even if a PK were dropped
Fact → dim orphan sweep	21	zero unmatched foreign keys across all Gold relationships (incl. nullable date keys, checked when present)
Silver → Gold set equality	10	no row lost or invented in either direction (catches key swaps row counts miss)
Value reconciliation	3	SUM(price) = 983,139,078.63; SUM(freight) = 239,045,494.22; SUM(payment_value) = 1,222,184,572.85 — silver = gold = baseline to the cent
Business rules	6	canceled → NULL delivery keys; delivered → delivery date present; reviews → delivered only; per-order payments = price + freight; fact_orders purchase key matches dim_order; delivery-band Unknown = canceled count
Fact → dim → silver key consistency	3	surrogate keys resolve back to the true business keys (added after incident #29.3)
dim_date coverage	1	every date key used anywhere lies inside the calendar (20190101–20270120)

Cross-layer financial identity (verified): per-order SUM(payment_value) = SUM(price + freight_value) on **all 1,000,000 orders with zero mismatches** — payments reconcile exactly with customer invoices.

Usage: run the whole file after any gold reload; every row must show PASS before refreshing Power BI.

8. Gold Layer — Dimensional Model

The Gold layer is a **Kimball-style fact constellation (galaxy schema)**: four fact tables at four declared grains, six conformed dimensions, one hub dimension (`dim_order`) that lets cross-process questions be answered without fact-to-fact joins. This shape is appropriate for Power BI because Power BI's filter propagation works relationship-by-relationship — a clean constellation gives every measure one unambiguous filter path.

Dimensions:

Table	Grain	Rows	PK	Business keys (UNIQUE)	Purpose
<code>dim_customer</code>	customer account (1 per order placement)	1,000,000	customer_key (IDENTITY)	customer_id	Two-level identity: customer_unique_id = person (279,199); segment, gender, age, ZIP, customer_type
<code>dim_product</code>	product	2,000	product_key (IDENTITY)	product_id	Category, brand, dimensions, unit cost, list price
<code>dim_seller</code>	seller	500	seller_key (IDENTITY)	seller_id	Seller company and contact attributes
<code>dim_geolocation</code>	ZIP prefix	900	zip_code_key (IDENTITY)	zip_code_prefix	City, state, centroid lat/lng
<code>dim_date</code>	day	2,942	date_key (yyyymmdd INT)	full_date (UNIQUE index)	Calendar: ISO weeks, iso_year (Thursday rule), day_of_week (Mon=1, DATEFIRST-independent), year_month, flags
<code>dim_order</code>	order (hub)	1,000,000	order_key (IDENTITY)	order_id	Status, 5 role-playing date keys, delivery_days_band, is_late

Facts:

Table	Grain	Rows	PK	FKs	Measures/attributes
<code>fact_sales</code>	order line	2,199,819	order_key + order_item_id	order, customer, product, seller, shipping date, purchase date	price (post-discount), freight_value, discount_rate
<code>fact_orders</code>	order (process measures)	1,000,000	order_key	order, customer, purchase date	delivery_days, days_to_approve, hours_to_approve, days_to_carrier
<code>fact_payments</code>	payment transaction	1,149,371	order_key + payment_sequential	order, customer, purchase date	payment_type, installments, payment_value
<code>fact_reviews</code>	review	933,748	review_id	order, customer, creation date	review_score

Business rules computed once in Gold:

- `delivery_days_band` (`dim_order`): Fast ≤ 3 days · Normal ≤ 7 · Slow ≤ 14 · Very Slow > 14 · Unknown (canceled) — CHECK-constrained.
- `is_late` (`dim_order`): 1 when actual delivery $>$ estimated; 0 on/before; NULL when undelivered.
- All five `dim_order` date keys (purchase, approved, carrier, delivered [nullable], estimated) and both `fact_sales` date keys (purchase, shipping) resolve to `dim_date` — the role-playing date dimension.

9. Grain Documentation

Grain is the analytical contract: every measure, relationship, and visual is only correct relative to the grain of the table it reads.

Gold table	Grain	Consequence
fact_sales	one row per order item	Revenue/units are line-additive; order counts require COUNT(DISTINCT order_key) or must read fact_orders
fact_orders	one row per order	Order counts and delivery averages live here; no revenue measures
fact_payments	one row per payment transaction	~149K orders have 2 payment rows — payment measures are transaction-grain; never summed as revenue
fact_reviews	one row per review	Score averages are review-grain; reviews exist only for delivered orders
dim_customer	one row per customer account	Person-level metrics require DISTINCTCOUNT(customer_unique_id)
dim_order	one row per order (hub)	Order attributes exist once; facts never join each other

How the hub prevents double counting and fact-to-fact joins: fact_sales (lines), fact_orders (order measures), fact_payments (payments), and fact_reviews (reviews) all carry order_key and connect to dim_order. "Do reviews differ by delivery band?" is answered fact_reviews → dim_order → GROUP BY delivery_days_band — never by joining two facts. Cross-fact comparisons (payments vs invoices) are made **after aggregating each fact to order grain**, then joined 1:1 — the pattern the validation suite enforces and documents.

10. Relationships & Semantic Model

Verified state (18 relationships = 15 active + 3 intentionally inactive, all Many-to-one, all single-direction):

Relationship	Cardinality	Direction	State
fact_sales → dim_customer / dim_product / dim_seller / dim_order / dim_date (purchase)	*:1	Single	Active
fact_sales → dim_date (shipping limit date)	*:1	Single	Inactive (role-playing reserve)
fact_orders → dim_customer / dim_order / dim_date (purchase)	*:1	Single	Active
fact_payments → dim_customer / dim_order / dim_date (purchase)	*:1	Single	Active
fact_reviews → dim_customer / dim_order / dim_date (creation)	*:1	Single	Active
dim_customer → dim_geolocation (ZIP)	*:1	Single	Active
dim_seller → dim_geolocation (ZIP)	*:1	Single	Inactive (avoids a second filter path to fact_sales)
dim_order → dim_date (purchase)	*:1	Single	Inactive (avoids a second date path to the facts)

How the model prevents analytical failure:

- **No fact-to-fact relationships** — cross-process questions route through conformed dimensions (the dim_order hub), so no visual can double-count or fan out.
- **All single-direction** — filters only flow dimension → fact; no ambiguous paths, no bidirectional loops (the project previously *observed* Power BI auto-deactivating relationships when 1:1 cardinality forced Both-direction filters — documented in §29 and fixed by forcing *:1).
- **One role-playing date dimension** with three active date paths (sales purchase, payments purchase, reviews creation) and two intentionally inactive ones — time intelligence works from one marked date table without ambiguous paths.
- **Keys hidden, business fields exposed** — technical columns (*_key, order_id) are hidden from the report view.

11. DAX Measures & KPI Dictionary

All 24 measures live in a dedicated _Measures table, organized in display folders (Sales / Profitability / Operations / Customers / Customer Experience / Payments). Definitions below are the implemented ones, verified by reading the .pbix.

Financial measures (all Delivered-only):

```
Total Revenue =
CALCULATE(
    SUM(fact_sales[price]) + SUM(fact_sales[freight_value]),
    dim_order[order_status] = "Delivered"
)
Net Revenue =
CALCULATE(SUM(fact_sales[price]), dim_order[order_status] = "Delivered")
Gross Profit =
CALCULATE(
    SUMX(fact_sales, fact_sales[price] - RELATED(dim_product[cost])),
    dim_order[order_status] = "Delivered"
)
Freight Collected =
CALCULATE(SUM(fact_sales[freight_value]), dim_order[order_status] = "Delivered")
Gross Profit Margin = DIVIDE([Gross Profit], [Net Revenue])
```

Volume and order measures:

```
Total Orders =
CALCULATE(COUNTROWS(fact_orders), dim_order[order_status] = "Delivered")
Total Orders All = COUNTROWS(fact_orders)
Canceled Orders =
CALCULATE(COUNTROWS(fact_orders), dim_order[order_status] = "Canceled")
Cancellation Rate = DIVIDE([Canceled Orders], [Total Orders All])
Canceled Order Value =
CALCULATE(
    SUM(fact_sales[price]) + SUM(fact_sales[freight_value]),
    dim_order[order_status] = "Canceled"
)
Units Sold =
CALCULATE(COUNTROWS(fact_sales), dim_order[order_status] = "Delivered")
```

```
AOV = DIVIDE([Total Revenue], [Total Orders])
```

Customer measures (delivered-based, person grain via customer_unique_id):

```
Total Customer =          -- persons with >= 1 delivered order
CALCULATE(
    DISTINCTCOUNT(dim_customer[customer_unique_id]),
    dim_order[order_status] = "Delivered"
)
ALL Persons = DISTINCTCOUNT(dim_customer[customer_unique_id])
Repeat Customers =        -- persons with >= 2 delivered orders
COUNTROWS(
    FILTER(
        VALUES(dim_customer[customer_unique_id]),
        CALCULATE(COUNTROWS(fact_orders), dim_order[order_status] = "Delivered") >= 2
    )
)
Repeat Purchase % = DIVIDE([Repeat Customers], [Total Customer])
Orders per Customer = DIVIDE([Total Orders], [Total Customer])
```

Experience, operations, payments, time:

```
Avg Review Score = AVERAGE(fact_reviews[review_score])
On-Time Rate =
VAR OnTimeOrders =
    CALCULATE(COUNTROWS(dim_order),
        dim_order[is_late] = FALSE, dim_order[order_status] = "Delivered")
RETURN DIVIDE(OnTimeOrders, [Total Orders])
Avg Delivery Days =
CALCULATE(AVERAGE(fact_orders[delivery_days]), dim_order[order_status] = "Delivered")
Avg Installments =
CALCULATE(AVERAGE(fact_payments[payment_installments]), dim_order[order_status] = "Delivered")
Payment Mix % =
DIVIDE(
    CALCULATE(SUM(fact_payments[payment_value]), dim_order[order_status] = "Delivered"),
    CALCULATE(SUM(fact_payments[payment_value]),
        dim_order[order_status] = "Delivered", ALL(fact_payments[payment_type]))
)
Revenue YoY % =
VAR CurrentRevenue = [Total Revenue]
VAR PrevRevenue = CALCULATE([Total Revenue], DATEADD(dim_date[full_date], -1, YEAR))
RETURN IF(ISBLANK(CurrentRevenue), BLANK(),
    DIVIDE(CurrentRevenue - PrevRevenue, PrevRevenue))
```

Why the definitions are shaped this way:

- **Delivered filter embedded in the base measures** — consistency never depends on page filters; "Total Orders" always means completed orders.
- **Revenue = price + freight; Net Revenue = price; Gross Profit = price – cost.** Freight is *customer-billed* revenue (verified: per-order payments reconcile exactly with price + freight) but is excluded from margin because no carrier-cost data exists — a declared business definition, not an assumption.
- **COUNTROWS vs DISTINCTCOUNT:** fact grain decides — fact_orders is one row per order (COUNTROWS), person counts need DISTINCTCOUNT on customer_unique_id.

- **SUMX + RELATED** is required only for Gross Profit, because unit cost lives in dim_product, not on the fact.
- **Time intelligence** (DATEADD, SAMEPERIODLASTYEAR patterns) is valid because dim_date is marked as the report's date table.

Reconciliation baselines (no filters, SQL-verified): Total Revenue 1,133,623,855.90 · Net Revenue 910,610,632.36 · Gross Profit 233,890,629.40 (margin 25.69%) · Freight Collected 223,013,223.54 · Units Sold 2,054,087 · Total Orders 933,748 delivered / 1,000,000 total · Canceled value 88,560,716.95 · Customers 276,017 · Repeat 226,611 (82.1%) · Avg Review 4.0348 · On-Time 90.14% · Avg Delivery 7.90 days · Avg Installments 3.95.

12. Business Questions

Every visual is titled as the business question it answers.

#	Business Question	Page	Visual	Dimensions	Measures
1	How is revenue trending over time?	1	Line/area by year_month	date (year_month)	Total Revenue
2	Is revenue growing year over year?	1	Column by year (+ YoY context)	date (year)	Total Revenue, Revenue YoY %
3	How much revenue do cancellations cost?	1	Area by year	date (year), status = Canceled	Canceled Order Value
4	Which categories generate the most revenue?	2	Horizontal bar	product category	Net Revenue
5	Which categories balance revenue and margin?	2	Scatter (log-log)	product category	Net Revenue, Gross Profit Margin
6	Which products drive the most revenue?	2	Table (Top 10)	product name	Net Revenue, Gross Profit, Margin
7	Do discounts increase sales?	2	Column + line combo	discount band	Net Revenue, Units Sold
8	Which categories keep the healthiest margins?	2	Column	product category	Gross Profit Margin
9	Which age groups generate the most revenue?	3	Column	age band	Net Revenue
10	What share of customers come back?	3	Donut	customer type	Total Customer, Repeat Purchase %
11	Which segments generate the most revenue?	3	Column	customer segment	Net Revenue
12	Which states generate the most revenue?	3	Shape map	state	Net Revenue
13	Does delivery speed affect satisfaction?	4	Column (hero)	delivery band	Avg Review Score
14	Do busy months slow delivery?	4	Column + line combo	month	Total Orders, Avg Delivery Days
15	Which states are slowest to deliver?	4	Area (ascending)	state	Avg Delivery Days
16	Which sellers generate the most revenue?	4	Horizontal bar	seller	Net Revenue
17	How do reviews change through the year?	5	Column + line combo	month	Review Count, Avg Review Score
18	How are customers paying?	5	Funnel	payment type	Payment Mix %

#	Business Question	Page	Visual	Dimensions	Measures
19	How much do customers finance?	5	Column	payment type	Avg Installments
20	How are review scores distributed?	5	Column	review score	Review Count

13. Power BI Dashboard Documentation

Model: Import mode; 10 tables imported from Gold (prefixed `gold` in the model); technical keys hidden; `dim_date` marked as date table; auto date/time disabled.

Page 1 — Executive Overview. *Audience:* leadership. *Decision supported:* trajectory, scale, and revenue leakage. KPI cards: Total Revenue (\$1.13B) · Total Orders (934K) · Total Customer (276K) · AOV (\$1,214) · Cancellation Rate (6.63%) · Canceled Order Value (\$88.6M). Visuals: revenue trend by year_month · revenue by year with growth-year coloring · canceled order value by year (red = lost money) · Year slicer · page-navigation button.

Page 2 — Sales & Profitability. *Audience:* commercial/finance. *Decision supported:* where revenue and profit come from, and whether discounting pays. Visuals: revenue by category (bar) · revenue-vs-margin scatter (log-log, bubble = gross profit) · top-10 products table · discount-band combo (revenue columns + units line) · margin-by-category columns (red = below-average margin) · 3 slicers (category, brand, discount band).

Page 3 — Customer & Geography. *Audience:* marketing/CRM. *Decision supported:* who the customers are, where they are, and how loyal they stay. Visuals: repeat-vs-one-time donut (delivered-based, center = 276K) · net revenue by age band · net revenue by segment · net revenue by state (single-hue map) · 4 slicers (segment, gender, age band, state).

Page 4 — Operations & Fulfillment. *Audience:* operations. *Decision supported:* fulfillment performance and its customer impact. Visuals: review score by delivery band (hero — Fast 4.4 / Normal 4.3 / Slow 4.1 / Very Slow 1.5) · orders + avg delivery days by month combo (peak months highlighted) · avg delivery days by state (ascending) · top sellers by net revenue · delivery-band slicer.

Page 5 — Payments & Reviews. *Audience:* payments/product. *Decision supported:* payment-rail strategy and experience trends. Visuals: payment mix funnel (credit_card 50.7% → boleto 1.2%) · review volume + score by month combo (December dip visible) · avg installments by payment type · review-score distribution · payment-type and Year slicers.

Navigation: NEXT/BACK action buttons chain the five pages in storytelling order.

14. Dashboard Design Decisions

Decision	Reason	Benefit	Trade-off
Every visual titled as a business question	Titles carry the analytical intent, not the chart type	A reviewer reads the story without orientation cost	Longer titles than the default field names
Dark navy theme, gold KPI values, white cards	Matches the project's documentation/deck identity; dark surfaces make semantic green/red readable	One coherent visual identity across 5 pages	Dark themes need contrast discipline

Decision	Reason	Benefit	Trade-off
Semantic green/red ONLY for true good/bad (margin, YoY, delivery cliff, score distribution); neutral single color for magnitude comparisons	Red on "65+ customers" or "VIP segment" would assert a problem that does not exist	Colors carry meaning; nothing is falsely alarming	Magnitude charts lose per-category color variety (accepted)
KPI cards (≥ 3 per page) with embedded delivered filters	Executives read state before trends	Consistent headline numbers on every page	Card values are context-dependent by design
Month-level charts use <code>month_short_name</code> (seasonality view) while Page 1 trend uses <code>year_month</code> (chronology)	Page 4/5 answer seasonal questions; Page 1 answers trajectory	Each granularity serves its question	<code>month_short_name</code> aggregates across years — chronology lives on Page 1
Data labels ON only for small-N charts; OFF for 85-point monthly trends	Labels collide at high cardinality	Readable trends; tooltips carry values	Exact values need hover on dense charts
Technical keys hidden; business fields exposed	Reviewers see business language, not surrogate keys	Cleaner field lists	None — keys remain available to the model
Power Query kept light; SQL is the transformation layer	All cleansing is scripted, re-runnable, and validated in SQL	One transformation language, one validation suite	Power Query is transport + light shaping only
DAX centralized in <code>_Measures</code> , display folders per theme	One definition per KPI; no visual-level calculations	Consistency; maintainability	None
Import mode	~7.4M rows is trivial for VertiPaq; sub-second visuals	Performance, full DAX feature set	Data freshness = refresh-time, acceptable for a static-period dataset
Validation suite gates every refresh	80 automated checks prove the model before it is consumed	No unverified numbers can reach the dashboard	Discipline required to run it

15. Seasonality & Time Analysis

Observed seasonality (stated pattern: May, July, August, November, December):

- **November is the revenue super-peak.** Monthly delivered revenue spikes every November and the spike grows with the business: ~\$27M (Nov 2019) → ~\$43M (2023) → ~\$54M (2024) → ~\$65M (Nov 2025), against a ~\$10–20M typical month.
- **November/December are the order-volume peaks.** Aggregated by calendar month, November (~200K delivered orders) and December (~135K) run 2–3× a typical month (~55–60K).
- **Review volume peaks in the same window:** December (~150K reviews) and November (~105K) are the highest review months; July/August (~95K) and May (~80K) are secondary elevated months.
- **Review scores peak in May (4.14) and July (4.13)** and dip in December (3.93) and January (3.94) — a recurring year-end dip visible in every year of the series.

Which metrics show it: order volume (Page 4 combo), revenue (Page 1 trend and the November spikes), review volume (Page 5 combo), and review scores (May/July highs, December lows).

What it affects: revenue and order capacity concentrate in Q4; customer-feedback volume and sentiment shift with it. Gross margin by month is not charted, so seasonality's *profit* effect is not established on the dashboard.

What cannot be concluded: the dashboard contains no marketing, promotion, holiday, or weather data, so the *cause* of the seasonality cannot be established from this project. The December score dip is also **not** explained by delivery slowness — December was the fastest delivery month of the year (7.77 days) — so the cause lies outside the measured operations data.

Business interpretation (supported): Q4 capacity, inventory, and staffing plans should be sized to a November peak that grows each year; the December experience dip should be investigated on its own merits (expectations, mix) rather than assumed to be a logistics failure.

16. Business Insights

Each insight: what happened → evidence → business meaning → possible action → confidence (High = exact, verified numbers; Medium = clear pattern, approximate read; Low = indicative only).

1. Electronics is 74% of revenue — at the lowest margin of any category

Insight: Electronics generates \$676M (74% of \$911M net revenue) at an 18.2% margin — the lowest of all seven categories. **Evidence:** Page 2 revenue bar + margin columns; verified category sums. **Business Meaning:** Company revenue is effectively a single-category business; blended margin (25.7%) is a weighted average dominated by electronics. **Possible Action:** Margin-aware pricing or procurement in electronics; deliberate growth of high-margin categories (fashion 62.5%, toys 55.3%). **Confidence: High** (exact verified sums).

2. Delivery speed has a satisfaction cliff at 15+ days

Insight: Avg review score holds at 4.4/4.3/4.1 for Fast/Normal/Slow, then collapses to 1.5 for Very Slow (15+ days). **Evidence:** Page 4 hero chart; 66K very-slow orders; score distribution shows the 1–2-star mass (161K reviews). **Business Meaning:** The SLA threshold is ~14 days — crossing it destroys the customer experience, not just dents it. **Possible Action:** Treat 14 days as an operational SLA; prioritize the very-slow cohort. **Confidence: High** (band definition and scores are exact; the *cause* of slowness is not established).

3. 2025 cancellations accelerated disproportionately

Insight: Canceled order value jumped to \$37M in 2025 — ~12% of order value vs a stable ~6% share in 2019–2024. **Evidence:** Page 1 canceled-value chart (red gradient, labeled by year). **Business Meaning:** Either the cancellation rate or the average canceled order size grew — the dashboard shows the value but not the split. **Possible Action:** Root-cause review of 2025 cancellations (reason, timing, category) before it becomes the new baseline. **Confidence: High** on the value jump; the driver is Not Available in this dataset.

4. The business halved in 2020–2021, then grew ~6×

Insight: Revenue collapsed from \$131M (2019) to \$52M (2020, –60%), stayed flat in 2021, then grew four consecutive years to \$309M (2025). **Evidence:** Page 1 year columns with YoY coloring (red = decline). **Business Meaning:** Growth is real and sustained but percentage growth is maturing (+50% → +33% → +18%). **Possible Action:** Plan for large absolute increments; investigate what distinguished 2022's recovery. **Confidence: High** on the numbers; the downturn's cause is Not Available in this dataset.

5. Retention is near-ceiling — the growth lever is acquisition and second-purchase conversion

Insight: 82.1% of delivered customers (226,611 of 276,017) are repeat customers; orders per customer = 3.38.

Evidence: Page 3 KPIs; delivered-based definition (delivered persons, ≥ 2 delivered orders = repeat).

Business Meaning: Loyalty is not the constraint — the ~49K one-time delivered customers are the conversion pool.

Possible Action: Second-purchase campaigns targeting the one-time cohort; acquisition investment for net-new growth.

Confidence: High on the counts; the definition is delivered-based and documented.

6. November is a scaling super-peak

Insight: November revenue spikes recur every year and grow: ~\$27M (2019) → \$65M (2025); November order volume (~200K) is 2–3× a typical month — and December fulfillment was the *fastest* of the year (7.77 days).

Evidence: Page 1 trend, Page 4 combo. **Business Meaning:** Q4 demand concentration is structural and growing; operations absorbed it without degradation. **Possible Action:** Formalize peak-season capacity planning; protect the December experience for reasons *other* than speed (see insight 8). **Confidence:** High.

7. 25–44-year-olds generate 59% of revenue; 65+ generates under 1%

Insight: 35–44 (\$272M) + 25–34 (\$265M) = 59% of net revenue; 55–64 (\$41M) and 65+ (\$6M) are marginal.

Evidence: Page 3 age-band chart. **Business Meaning:** The customer base (and its spending) is concentrated in working-age adults; 55+ underperformance is unexplained by the dashboard. **Possible Action:** Anchor acquisition on 25–44; one diagnostic pass on the 55+ gap before writing the segment off. **Confidence:** High on the revenue split; the *why* is Not Available.

8. Discounts show no measurable volume lift

Insight: ~97.5% of units and revenue are non-discounted (\$888M); the three discounted bands combined contribute ~\$23M and ~190K units, with flat unit counts per band. **Evidence:** Page 2 discount-band combo. **Business Meaning:** Current discounting (10–30% on ~8.6% of lines) adds margin pressure without observable volume response. **Possible Action:** Review the discount program's purpose; test discounts where elasticity might exist rather than broadly. **Confidence:** Medium (correlation; no price-elasticity model in scope).

9. Payment value concentrates on card rails, and financing is card-exclusive

Insight: credit_card carries 50.7% of payment value (debit_card 7.2%); average installments are 6.4/6.3 on cards vs exactly 1.0 for every other type. **Evidence:** Page 5 funnel + installments chart; verified per-order payment reconciliation. **Business Meaning:** Card processing economics are material at \$1.22B payment volume; non-card financing is an untested option. **Possible Action:** Negotiate card fees; pilot installments/BNPL for wallet users. **Confidence:** High on the distribution; the *lift* from extending financing is untested.

10. Geography concentrates revenue but not delivery performance

Insight: A handful of states (CA, TX, NY, FL darkest) carry most revenue, yet delivery time is statistically flat across all states (7.88–7.95 days). **Evidence:** Page 3 map + Page 4 state chart. **Business Meaning:** Where customers are matters for revenue; where they live does not affect how fast they are served. **Possible Action:** Focus regional marketing on proven states; delivery investment should target the process, not locations. **Confidence:** High on the flat delivery spread; revenue concentration is exact.

11. Consumer is 70% of revenue; VIP is 10%

Insight: Consumer \$638M, Corporate \$182M, VIP \$91M of \$911M net revenue. **Evidence:** Page 3 segment chart. **Business Meaning:** The consumer business funds the company; Corporate and VIP are meaningful but secondary. **Possible Action:** Segment-specific playbooks; verify VIP's margin profile (not charted) before investing. **Confidence:** High on the split.

12. The 2-star review bulge is an anomaly worth investigating

Insight: The score distribution is non-monotonic: 2-star (115K) exceeds 3-star (69K). Most review distributions decline monotonically from 5-star. **Evidence:** Page 5 score distribution chart. **Business Meaning:** A distinct driver may exist behind exactly-2-star reviews (different from the 1-star very-slow cohort). **Possible Action:** Read 2-star review comments (available in Silver) to identify the driver. **Confidence:** Medium — the bulge is verified; its cause is not.

17. Executive-Level Insights

- **Revenue:** \$309M in 2025, four consecutive growth years after a 2020–21 collapse (\$52M/yr); November alone reached \$65M in 2025 — a recurring, scaling super-peak.
- **Profitability:** \$234M gross profit at a 25.7% margin — but the margin is a blend: the revenue engine (electronics, 74% of revenue) runs at the lowest margin (18.2%), while fashion earns 62.5% on 7% of revenue.
- **Customers:** 276K delivered customers, 82% of them repeat — retention is near-ceiling; the growth lever is acquisition and converting the ~47K one-time buyers.
- **Concentration:** 74% of revenue is electronics; 70% is Consumer; 59% comes from ages 25–44; a handful of states dominate the map. Concentration is the dashboard's most consistent structural finding.
- **Operations:** 90.14% on-time, 7.9-day average delivery, flat across all states, and *faster* in the December peak (7.77 days) — operations scale with demand.
- **Experience:** satisfaction collapses from 4.1 to 1.5 when delivery crosses 15 days — the single largest experience finding — with a secondary December dip that delivery speed does not explain.
- **Payments:** credit_card carries 50.7% of \$1.22B payment value; financing (6.4/6.3 installments) exists only on cards. Cancellations cost \$88.6M lifetime, with 2025 the worst year (\$37M).
- **Seasonality:** May, July, August, November, and December are the elevated months across volume and review activity; December combines peak feedback volume with the year's lowest scores.

18. Customer Analysis

Identity model: the source models identity at two levels — `customer_id` (one account per order placement, 1M) and `customer_unique_id` (the person, 279,199). Person-level metrics use `DISTINCTCOUNT(customer_unique_id)`; conflating the two levels would inflate customer counts ~3.6×.

Segments (net revenue, delivered): Consumer \$638M (70%), Corporate \$182M (20%), VIP \$91M (10%).

Loyalty (delivered-based definition, locked for this project):

Metric	Value	Definition
Delivered customers	276,017	persons with ≥ 1 delivered order
Repeat customers	226,611	persons with ≥ 2 delivered orders

Metric	Value	Definition
Repeat Purchase %	82.1%	repeat / delivered customers
One-time (delivered)	49,406	persons with exactly 1 delivered order
Orders per Customer	3.38	delivered orders / delivered customers

Interpretation: an 82% repeat rate is exceptionally high — in this dataset, once a customer completes a delivery, they almost certainly return. Two implications: (a) retention programs have little headroom; (b) the ~49K one-time delivered customers are the acquisition-conversion frontier. The all-orders variant of the same analysis (84.1% repeat) exists in the SQL layer; the delivered-based definition is the project's locked choice.

Caveats: the dataset is synthetic, so the 82% level should be presented as a demonstration of method. The customer_type column on dim_customer classifies at account grain (Repeat = person placed ≥ 2 orders, any status) and is *not* the same basis as the delivered person measures — documented so the two are never mixed on one page.

19. Product & Sales Analysis

Category performance (net revenue / gross profit / margin, delivered):

Category	Net Revenue	Gross Profit	Margin
electronics	\$676M	~\$123M	18.2%
furniture	\$102M	~\$42M	40.7%
fashion	\$64M	~\$40M	62.5%
home_goods	\$32M	~\$16M	49.2%
auto	\$20M	~\$5M	25.7%
toys	\$12M	~\$7M	55.3%
books	\$5M	~\$2M	39.9%

- **Concentration:** electronics = 74% of net revenue; the top category earns the lowest margin — the central tension of the category mix.
- **Products:** the top 10 products each contribute \$4.38–4.56M (~5% of revenue combined); product_name-level analysis is avoided on the dashboard because product cardinality is high — category is the working granularity, with the top-10 table as the drill-down.
- **Discounts:** 21 distinct rates (10%–30%) apply to ~8.6% of lines; discounted bands contribute ~2.5% of revenue and units, with no volume lift visible across bands (correlation only — no elasticity model in scope).
- **Freight:** freight_value is customer-billed revenue (verified: per-order payments = price + freight exactly). It is included in Total Revenue and excluded from Gross Profit by business definition (no carrier-cost data exists).

20. Geography Analysis

- **Customer geography:** revenue concentrates in a handful of states — California and Texas are the darkest on the map; most states are near-zero. The map uses a single-hue intensity ramp (light → dark = low → high revenue) rather than a red/green divergence, because a small state is not "bad".
- **Seller geography:** sellers carry their own ZIP attribute; seller location is not joined to customer location anywhere in the model (an order spans sellers from multiple states), so "where sellers are" is documented, not analyzed as a revenue driver.
- **Delivery by geography:** every state delivers in 7.88–7.95 days — geography does **not** differentiate delivery performance (Page 4). Operations is a process property, not a location property.
- **Implication:** regional marketing should follow proven demand (the dark states); delivery investment should target the fulfillment process, not specific states.

21. Operations & Delivery Analysis

Status model: 1,000,000 orders = 933,748 delivered (93.37%) + 66,252 canceled (6.63%). Canceled orders have structurally NULL delivery timestamps and keys — preserved, never imputed.

Delivery performance (delivered orders):

- **On-Time Rate: 90.14%** — `is_late` = FALSE among delivered orders (late = actual delivery after the estimated date).
- **Average delivery: 7.90 days** (purchase → customer delivery), with a per-state spread of only 7.88–7.95 days — geography does **not** differentiate fulfillment.
- **December delivery: 7.77 days — the fastest month of the year**, despite December carrying peak order volume. Peak-season operations scale with demand.

Delivery bands (dim_order, CHECK-constrained):

Band	Definition	Avg Review Score
Fast	≤ 3 days	4.4
Normal	4–7 days	4.3
Slow	8–14 days	4.1
Very Slow	> 14 days	1.5
Unknown	canceled (NULL)	—

The satisfaction cliff: scores are stable through "Slow" and collapse at "Very Slow" — a 2.6-point drop. ~66K orders sit in the Very Slow band. The dashboard establishes the *association* between the band and the score; it cannot prove duration *causes* the score (no controlled comparison exists in observational data).

Freight: freight_value is billed to the customer and included in Total Revenue; it is excluded from Gross Profit by business definition (no carrier-cost data exists to net it against).

22. Review & Customer Experience Analysis

- **Volume:** 933,748 reviews — one per delivered order (100% coverage of delivered orders; canceled orders are never reviewed). Review volume peaks in December (~150K) and November (~105K).
- **Average:** **4.03** (on a 1–5 scale), stable at 4.02–4.14 across Feb–Nov.
- **Distribution:** 5-star 469K · 4-star 234K · 3-star 69K · **2-star 115K** · 1-star 46K — the non-monotonic 2-star bulge (2-star > 3-star) is the distribution's anomaly.
- **The December dip:** scores fall to 3.93 in December — the year's lowest — while December *delivery* was the fastest month of the year. The dashboard therefore **rules out delivery slowness** as the December explanation and points to unmeasured year-end factors. No causal claim is made.
- **The delivery relationship:** the score-by-band cliff (4.4 → 1.5) is the strongest experience finding. The dashboard establishes the *association* between delivery duration and scores; causality is not established (no controlled comparison exists in observational data).
- **Coverage note:** because reviews exist only for delivered orders, review metrics are inherently a delivered-order experience measure — no missing-data adjustment is needed or made.

23. Payment Analysis

- **Grain:** one row per payment transaction (`order_id` + `payment_sequential`); ~149K orders were paid with two transactions.
- **Types (share of delivered payment value):** `credit_card` 50.7% · `paypal` 15.8% · `apple_pay` 9.7% · `bank_transfer` 7.7% · `voucher` 7.5% · `debit_card` 7.2% · `boleto` 1.2%.
- **Financing:** average installments = 6.4 (`credit_card`) and 6.3 (`debit_card`); all non-card types sit at 1.0 — financing is exclusively a card feature in this dataset.
- **Reconciliation (verified):** per-order $\text{SUM}(\text{payment_value}) = \text{SUM}(\text{price} + \text{freight_value})$ on **all 1,000,000 orders, zero mismatches** — payments reconcile exactly with customer invoices, including canceled orders.
- **Why no `dim_payment`:** payment attributes (`type`, `installments`) are transaction-grain facts, not descriptive dimensions — a `dim_payment` would add a join without adding analysis. Payment *analysis* happens by grouping `fact_payments` directly, which the model supports through its active relationships.
- **Modeling guardrail:** `payment_value` is never summed as *revenue* (revenue lives in `fact_sales`); the two reconcile per order, which the validation suite checks — a QA result, not a modeling invitation.

24. Financial Analysis

Definitions (locked, implemented exactly):

Measure	Formula	Delivered filter	Purpose
Total Revenue	SUM(price + freight_value)	Yes	What customers paid (invoices)
Net Revenue	SUM(price)	Yes	Merchandise revenue — the margin base
Gross Profit	SUM(price – product cost)	Yes	Merchandise profit
Gross Profit Margin	Gross Profit / Net Revenue	Yes	Profitability rate
Freight Collected	SUM(freight_value)	Yes	Shipping billed to customers
Canceled Order Value	SUM(price + freight_value)	Canceled	Revenue leakage monitor

Why freight is inside Revenue but outside Profit (business definition, not an assumption): per-order payments reconcile exactly with price + freight (verified on all 1M orders), proving freight is *billed to the customer*. No carrier-cost data exists, so freight cannot be netted against a known cost — it is therefore excluded from Gross Profit by declared definition rather than presented as a proven cost recovery.

Discounts: the transaction price is already post-discount (verified against list price); discount_rate is retained as an analysis attribute. Discount *value* per line = list price – transaction price (exact to the cent).

Canceled orders: excluded from every revenue/profit KPI (the delivered filter), monitored separately through Cancellation Rate (6.63%) and Canceled Order Value (\$88.6M, of which \$37M in 2025 alone).

Verified financial baseline (no filters): Total Revenue \$1,133,623,855.90 · Net Revenue \$910,610,632.36 · Gross Profit \$233,890,629.40 · Freight Collected \$223,013,223.54 · Canceled Order Value \$88,560,716.95. Cross-check: Total Revenue + Canceled Order Value = total payment_value (\$1,222,184,572.85) exactly.

25. Analytical Limitations

Data limitations

- **Synthetic dataset** — distributions are smooth and idealized (82% repeat, near-identical top sellers); patterns demonstrate method, not market reality.
- **Static period** — 2019–2025 snapshot; no ongoing ingestion.
- **No carrier-cost data** — freight cannot be netted into profitability by definition.
- **No refunds/returns data** — canceled-order payment handling is not observable.
- **No marketing, campaign, traffic, or inventory data** — acquisition, elasticity, and stockout effects cannot be analyzed.

Modeling limitations

- **Full-refresh architecture** — every layer reloads completely; no incremental loads.
- **No SCD** — dimensions are current-state; attribute history is not tracked.
- **dim_customer at account grain** — person metrics require DISTINCTCOUNT on customer_unique_id (documented, deliberate).

- **Geolocation aggregated to ZIP centroids** in Silver — coordinates are averages, not exact points.
- **Power BI Import mode** — data freshness equals refresh time; no incremental refresh.

Analytical limitations

- **Correlation $\neq$ causation** — the delivery-band/score cliff is an association; no controlled comparison exists.
- **The repeat definition is delivered-based** — a project choice (all-orders basis would read 84.1%); documented, not universal.
- **December score dip is unexplained** — delivery slowness is ruled out; the driver is outside the dataset.
- **Category margin analysis has no cost granularity** beyond unit cost — opex, marketing, and carrier costs are out of scope.

Operational limitations

- **No version control** — the project is a local folder (deliberate current state; recommended before publication).
- **Fail-loud loads without quarantine** — one bad row aborts a load.
- **Manual refresh workflow** — validation suite must be run before every Power BI refresh.

26. Reproducibility

Prerequisites: SQL Server 2017+ (uses `FORMAT= 'CSV' BULK INSERT` and `DROP INDEX IF EXISTS`), SSMS, Power BI Desktop, the 8 source CSVs in `\data\`.

1. **Database + schemas:** run `sql/bronze/create_database.sql`, then `sql/bronze/create_schema.sql` (bronze, silver, gold).
2. **Bronze tables:** `sql/bronze/create_bronze_tables.sql`.
3. **Bronze load:** edit the absolute CSV paths in `sql/bronze/load_bronze.sql` to `\data\...`, then execute.
4. **Silver:** `sql/silver/create_silver_tables.sql`, then `sql/silver/load_silver.sql`.
5. **Gold:** `sql/gold/create_gold_tables.sql`, then `sql/gold/load_gold.sql` (dims first, then facts — the script is ordered and includes the `fact_orders` date key, index, and FK).
6. **Validation:** `sql/validation/validate_gold.sql` — all 80 checks must show PASS.
7. **Analytical queries:** `sql/analytics/business_analysis.sql` (the ten business questions).
8. **Power BI:** open `PowerBI/US-E-Commerce_2.pbix`, update the SQL Server source to your instance, refresh.
9. **Semantic model:** relationships, hidden keys, and the date table are pre-configured and verified.
10. **Measures:** 24 measures are pre-built (§11).
11. **Dashboard:** 5 pages per §13.
12. **Refresh workflow:** re-run steps 4–7, confirm 80/80 PASS, then refresh Power BI.

27. Project Folder Structure

Actual structure on disk:

```

US E-Commerce Analytics – End-to-End Data Warehouse & BI/
├─ data/                                # 8 source CSVs (~857 MB)
├─ docs/
│   ├── DW_Architecture.drawio          # end-to-end pipeline diagram
│   ├── gold_layer.drawio / _v2 / _v3   # gold model evolution (v3 = current)
│   ├── gold_layer_v3.jpg               # exported model diagram
│   ├── PK and FK for tables.drawio
│   ├── PROJECT_DOCUMENTATION.md        # this file
│   ├── PROJECT_DOCUMENTATION.pdf       # PDF export
│   ├── archive/                        # superseded documentation versions
│   └─ (*.bkp drawio backups – editor artifacts)
├─ sql/
│   ├── bronze/                         # create_database, create_schema,
│   │   │                               # create_bronze_tables, load_bronze
│   ├── silver/                         # create_silver_tables, load_silver,
│   │   │                               # prepare_silver_tables (scratch)
│   ├── gold/                           # create_gold_tables, load_gold,
│   │   │                               # gold_layer_setup (scratch), gold.slnx
│   ├── analytics/                      # business_analysis.sql (Q1-Q10)
│   └─ validation/                      # validate_gold.sql (80 checks)
├─ PowerBI/
│   ├── US-E-Commerce.pbix              # earlier build
│   ├── US-E-Commerce_2.pbix            # final build (verified)
│   ├── US-E-Commerce-theme-v2.json     # importable theme
│   └─ US-E-Commerce-theme.json        # superseded v1
└─ WebDashboard/                       # Flask + Chart.js companion
    ├── app.py                           # live SQL-backed API
    ├── templates/index.html
    └─ static/chart.umd.min.js

```

Not present (current state): git repository, README.md, notebooks — a known, documented state (§29/§31).

28. Key Architecture & Engineering Decisions

Decision	Reason	Benefit	Trade-off
SQL Server 2025 as the warehouse engine	Set-based T-SQL transformations are scripted, testable, and re-runnable; SSMS ecosystem	One transformation language; 80-check validation suite	Requires SQL skilling (accepted — it is the target skill)
Medallion layering (Bronze/Silver/Gold)	Separates landing, cleansing, and modeling concerns	Re-derivable pipeline; failures isolated per layer	Three load scripts to maintain
Typed Bronze with fail-loud loads	The source is a known clean extract; bad data should abort, not quarantine	No silent corruption; simple recovery (fix + rerun)	One bad row aborts a load (accepted for a static source)
No Bronze audit columns	Single static source + full reload makes them valueless here	Simpler load	Rebuild lineage lives in this documentation instead

Decision	Reason	Benefit	Trade-off
Geolocation deduplicated in Silver (11,500 → 900 ZIP centroids)	The analytical model needs one row per ZIP; raw detail stays in Bronze	No revenue fan-out in Gold/BI	Coordinate precision reduced to ZIP centroid
Kimball fact constellation with a dim_order hub	Four process grains share order attributes; Power BI needs one filter path per measure	Zero fact-to-fact joins; no ambiguous paths	dim_order adds one table to maintain
Surrogate keys + hidden technical columns	Business keys remain for traceability; model exposes business language	Clean UX; stable relationships	IDENTITY keys require fact reloads after dim rebuilds (see §29.3)
dim_date built from the data (min → max event date + 1 year)	Deliveries and reviews occur after purchase; the calendar must cover every event date	No missing date keys (verified: max used key 20260120 < max calendar 20270120)	Calendar regenerates on reload
DATEFIRST-independent weekday + ISO year in dim_date	Weekday numbers and ISO week/year must not depend on session settings	Stable weekly analysis across environments	None
Index-after-load pattern (ix_fact_orders_order_purchase_date_key)	Sequential inserts during load, sorted index build after	Fact_orders load: 6m 41s → 5.7 s (~70×)	Index build time after load
Power BI Import mode	~7.4M rows is small for VertiPaq; full DAX feature set	Sub-second visuals; full time intelligence	Data freshness = refresh-time, acceptable for a static-period dataset
Validation suite gates every refresh	80 automated checks prove the model before consumption	No unverified number can reach the dashboard	Discipline to run it
Validation gates include silver→gold key consistency	A dimension reload reassigns IDENTITY keys; facts can go stale while all other checks pass	The scramble incident (§29.3) is now auto-detected	+3 checks to maintain
Web companion reads Gold live (Flask + SQL)	Demonstrates the Gold layer as an open contract beyond Power BI	Second consumer, zero model changes	Second artifact to run

29. Data Quality Issues & Fix History

#	Issue	Layer	Impact	Root Cause	Fix	Validation
1	Money columns typed FLOAT in early Bronze drafts	Bronze	Binary rounding noise on revenue sums	Initial DDL used FLOAT for money	All money columns DECIMAL(10,2); verified against source sums	Exact reconciliation (§7.5)
2	ROWTERMINATOR = '\n' rejected by the SQL Server build	Bronze	Loads failed	LF terminator must be passed as the hex literal	'0x0a' + FORMAT='CSV' + FIELDQUOTE — verified working	Row counts match CSVs exactly
3	Silver geolocation had no primary key	Silver	Deduped ZIP grain unenforced	PK omitted in the first DDL	pk_geo PRIMARY KEY (zip_code_prefix)	900 rows, unique

#	Issue	Layer	Impact	Root Cause	Fix	Validation
4	Silver DDL drift (trailing comma in geolocation block; Bronze DDL drifted from executed version)	Silver/Bronze	Saved scripts could not reproduce the database	Edits executed in SSMS but not saved back	DDL files corrected to match the live database	Scripts now reproduce the DB top-to-bottom
5	dim_date.full_date unindexed during fact loads	Gold	fact_orders load took 6m 41s (5M unindexed date lookups)	Join predicate on CAST(timestamp AS DATE) needed a lookup index	CREATE UNIQUE INDEX ix_dim_date_full_date	Load time 6m 41s → 5.7 s (~70×); physical reads 0
6	Customer surrogate-key scramble	Gold	999,987 fact rows joined to the wrong customer after dim_customer was rebuilt (customer_type added) — IDENTITY keys reassigned in a different order while facts kept old keys; row counts, orphans, and grain checks all still passed	Dimension reload without fact reload	Re-ran the four fact loads (facts re-resolve keys from current dims); verified 0 mismatches	New suite checks: fact → dim → silver key consistency (customer/product/seller)
7	validate_gold.sql payment-reconciliation check produced 697,368 false mismatches	Validation	The check joined two facts at row level before aggregating (lines × payments fan-out)	Grain error in the <i>check itself</i> , not the data	Rewritten to aggregate each fact to order grain first, then join 1:1	0 true mismatches on all 1M orders
8	Power BI 1:1 auto-detection forced bidirectional filters → ambiguous paths → Power BI silently deactivated the dim_order relationships	Power BI	dim_order cut off from all facts; delivered filters could not propagate	Three true 1:1 relationships (1 account = 1 order) forced Both-direction filters, creating a filter cycle	Forced *:1 Single on all relationships; re-activated; verified 18-relationship graph	Relationship dump verified post-fix
9	Theme JSON v1 rejected by Power BI validation	Power BI	Property groups nested inside the wildcard <code>/**/0</code> selector	Incorrect theme structure	v2 mirrors the shipped base-theme structure (named third-level keys); validated key-by-key	Imports cleanly
10	Dashboard hero labels read 44/43/41 instead of 4.4/4.3/4.1	Power BI	Data-label precision not set	Cosmetic but misleading on the hero visual	Label precision set to 1 decimal	Verified in render
11	Page 3 donut (customer_type basis) disagreed with the Repeat Purchase % card (82.93%/273K vs 82.19%/276K)	Power BI	Two definitions of one KPI on one page	Donut used the account-grain customer_type classification; card uses delivered-based person measures	Donut reconciled to the delivered-based card values in the final build	Both now read 82%/276K

30. Final Project Assessment

Data modeling. The Gold layer is a correctly grained fact constellation: four facts at four declared grains, six conformed dimensions, a hub dimension that eliminates fact-to-fact joins, surrogate keys with unique natural keys, and business rules computed once in dimensions rather than re-derived per visual. Physical PKs enforce every declared grain.

SQL architecture. The pipeline is fully scripted and re-runnable end to end: typed BULK INSERT landing, constraint-enforced conformance, dimension/fact loads with index-after-load patterns, and a date dimension built dynamically from the data itself (with the DATEFIRST-independent weekday and ISO-year logic that weekly analysis requires).

Data quality. The 80-check validation suite covers row counts, grain, orphans, set equality, value reconciliation, business rules, key consistency, and calendar coverage — and demonstrably works: it caught both a real grain error in its own reconciliation logic and a 999,987-row surrogate-key scramble in the warehouse.

Analytical correctness. Every KPI definition is locked, implemented once, and reconciled to SQL baselines. Revenue, margin, repeat, and delivery metrics follow documented, deliberate business definitions (delivered filter; freight inside revenue, outside profit; delivered-based repeat).

Power BI semantic model. Verified by direct file inspection: 24 measures in display folders, 18 single-direction relationships (15 active + 3 intentional), a marked date table, hidden technical keys, zero fact-to-fact paths, zero ambiguity.

Business usefulness. The dashboard answers ten business questions with verified numbers, and the analysis surfaces actionable findings: the very-slow delivery cliff (1.5-star), the 2025 cancellation acceleration (~12% of order value), the electronics margin tension (74% of revenue at the thinnest margin), and the seasonal November super-peak scaling to \$65M.

Maintainability. Everything is scripted; the pipeline rebuilds in minutes; the validation suite re-certifies it. The known maintenance debts: no version control, two scratch SQL files, and a page-size inconsistency (see §31).

31. Future Improvements

Short-term

- Version-control the project (git init + .gitignore for `data/`, IDE caches, backups) and add a README.
- Unify page canvas sizes to 1920×1080 across all five report pages.
- Add Review Count-style supporting measures where pages need them; finalize hidden-column and sort-by-column settings.
- Clean superseded scratch SQL files (`gold_layer_setup.sql`, `prepare_silver_tables.sql`) into an archive folder.

Medium-term

- Scheduled refresh with the validation suite as a pre-refresh gate (SQL Agent job that aborts on FAIL).
- Incremental loading using load timestamps (requires adding Bronze audit metadata).
- SCD Type 2 on `dim_customer/dim_product` if attribute history becomes relevant.
- Cohort retention analysis on `customer_unique_id` (the identity model already supports it).
- Category margin deep-dive: unit-cost granularity and freight-cost acquisition to make margin fully loadable.

Long-term

- CI/CD for the SQL pipeline (deploy scripts + automated validation on commit).
- Power BI deployment pipeline and workspace strategy; PBIP format for source control.
- Anomaly detection on the monthly revenue and review series.
- Forecasting on the seasonal revenue series (the November super-peak is the obvious candidate).
- Customer analytics extension: RFM and cohort analysis on the two-level identity model.

32. Final Executive Summary

1. **What was built:** an end-to-end analytics product — a three-layer SQL Server warehouse (Bronze → Silver → Gold) over 1M e-commerce orders, an 80-check automated validation suite, a verified Power BI semantic model with 24 governed measures, a five-page dashboard, and a live SQL-backed web companion.
2. **What data was used:** eight synthetic U.S. e-commerce extracts (2019–2025, ~6.3M raw rows): customers, orders, order items, payments, reviews, products, sellers, and geolocation.
3. **How it was transformed:** typed raw landing (Bronze), standardized and constraint-enforced conformance with one documented grain change (geolocation 11,500 → 900 ZIPs) and fail-loud quality policy (Silver), then reshaped into a Kimball fact constellation with surrogate keys, a role-playing date dimension, and pre-computed business rules (Gold).
4. **How the warehouse was designed:** a Kimball fact constellation — four facts at four grains (order line, order, payment transaction, review) around six conformed dimensions, with a `dim_order` hub that removes every fact-to-fact join, all verified by an 80-check suite that includes silver→gold key consistency.
5. **How Power BI was structured:** Import mode over the ten Gold tables, 18 single-direction relationships (15 active, 3 intentional), a marked date table, hidden technical keys, 24 centralized DAX measures reconciled to the SQL layer to the cent, and five question-titled dashboard pages in storytelling order.
6. **The most important KPIs:** Total Revenue \$1.13B (delivered, freight-inclusive) · Net Revenue \$911M · Gross Profit \$234M (25.7%) · 934K orders at a 6.63% cancellation rate (\$88.6M canceled value, \$37M in 2025 alone) · 276K delivered customers at an 82% repeat rate · 90.1% on-time delivery at 7.9 days average · 4.03 average review score.
7. **The major business insights:** electronics is 74% of revenue at the lowest margin (18.2%) — the mix-vs-margin tension; satisfaction collapses from 4.1 to 1.5 when delivery crosses 15 days; November is a scaling super-peak (\$27M → \$65M); 2025 cancellations doubled their historical share; discounts show no volume lift; the customer base is 82% repeat — growth is an acquisition problem.
8. **The main limitations:** synthetic data (patterns demonstrate method, not market reality); no cost granularity beyond unit cost, no marketing or carrier data; correlation is not causation (the delivery-satisfaction cliff is observational); full-refresh architecture without history tracking.

9. **Decisions this project supports:** category pricing and procurement (margin-aware), discount-program review, peak-season capacity planning, a 14-day delivery SLA, payment-rail economics, acquisition targeting on the 25–44 consumer core, and one-time-buyer conversion.
10. **What could be done next:** version control and README; scheduled refresh gated by the validation suite; incremental loads; SCD where history matters; cohort/RFM analytics on the two-level customer identity; forecasting on the seasonal revenue series.

End of documentation. The implementation is the source of truth; this document describes it as verified on 2026-09-26.